

StackFix Repair Management System

Complete User Flow & Feature Specification

Version 2.0 · Kigali, Rwanda · May 2026

OVERVIEW

What is StackFix?

StackFix is a repair management system designed specifically for repair businesses in Kigali, Rwanda. It enables technicians to log customer repair jobs, generate invoices, collect payments via USSD, and automatically track device status throughout the repair lifecycle — all from one dashboard.

This document describes the complete user flow including all new features: invoicing, USSD payment integration, automated status updates, AI assistance, and payment history management.

COLOUR KEY



Admin / Technician



Customer action



Automated / System



AI-assisted

SYSTEM ROLES

Who uses StackFix?

Admin

- Full access to all features
- Manages technician accounts
- Views invoices and payment history
- Exports reports and data
- Configures app settings and language

Technician (User)

- Logs in with credentials set by admin
- Creates repair tickets for customers
- Raises and sends invoices
- Updates ticket status as work progresses
- Views their own ticket history

Step-by-Step Process

The flow below covers the full lifecycle of a repair job from customer arrival to device pickup, including both payment models supported by StackFix.

- 1 Customer arrives at the shop** *[Customer]*
A customer walks in with a device they need repaired. No system action required at this point.



- 2 Technician opens a new repair ticket** *[Technician]*
The technician opens StackFix and creates a new repair ticket, collecting the following details from the customer:
 - Customer full name
 - Phone number (used for WhatsApp / SMS notifications)
 - Device type and model
 - Description of the fault or issue

AI AI Assist: The AI can auto-suggest the device type and common fault descriptions based on partial input, reducing typing time for the technician.



- 3 Repair ticket is created — status: Pending** *[System (Automated)]*
Once the technician submits the form, the system automatically creates the ticket and sends a notification to the customer.
 - Ticket assigned a unique ID (e.g. #007)
 - Initial status set to: Pending
 - Customer receives WhatsApp or SMS: “Your device has been received and is awaiting payment / service.”



- 4 Technician raises an invoice** *[Technician]*
Immediately after creating the ticket, the technician raises an invoice for the repair job. The invoice includes:
 - Customer name and contact
 - Device type and fault description
 - Line items: parts used, labour charge, any additional fees
 - VAT / tax (18% where applicable)
 - Total amount due
 - USSD payment code and instructions
 - Business name, address, and contact

AI AI Assist: The AI can pre-populate invoice line items based on the fault type and device model — e.g. if the fault is “screen crack” on a Samsung A54, it suggests “Screen replacement + Labour” with estimated amounts.



Invoice sent to customer [System (Automated)]

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The system sends the invoice directly to the customer via WhatsApp and/or SMS.

- Invoice includes a “Pay Now” button or USSD code (e.g. *182*1*amount#)
- Customer can view and pay directly from their phone
- Works on both smartphones and basic feature phones



At this point, the flow branches based on the business’s payment model:

PAYMENT MODEL — Choose one of two workflows

Model A — Pay Before Service

- Customer pays via USSD before repairs begin
- System receives payment confirmation automatically
- Ticket status auto-updates: Pending → Under Repair
- Technician is notified and begins work
- Best for: businesses requiring upfront payment

Model B — Pay on Pickup

- Service begins immediately after ticket is created
- Ticket status: Pending → Under Repair (manual by technician)
- Payment is collected when customer comes to pick up device
- Invoice is settled at the counter
- Best for: businesses with trusted customers or walk-ins



Device under repair [Technician]

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The technician works on the device. The ticket remains in Under Repair status throughout this phase.

- Technician can add notes or update parts used
- Ticket visible on dashboard with current status



Technician marks repair as complete [Technician]

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Once the device is fully repaired, the technician updates the ticket status.

- Status changes from: Under Repair → Completed
- Triggers automatic notification to customer



Customer notified — device ready for pickup *[System (Automated)]*

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The system automatically sends a notification to the customer.

- WhatsApp or SMS: “Your device is repaired and ready for pickup.”
- If Model B (pay on pickup): message also includes invoice and USSD payment link



Customer collects device *[Technician + Customer]*

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The customer comes to the shop to collect their device.

- For Model B: customer pays at the counter via USSD
- Payment confirmed automatically by the system
- Technician updates ticket status: Completed → Picked Up



Invoice marked paid — records updated *[System (Automated)]*

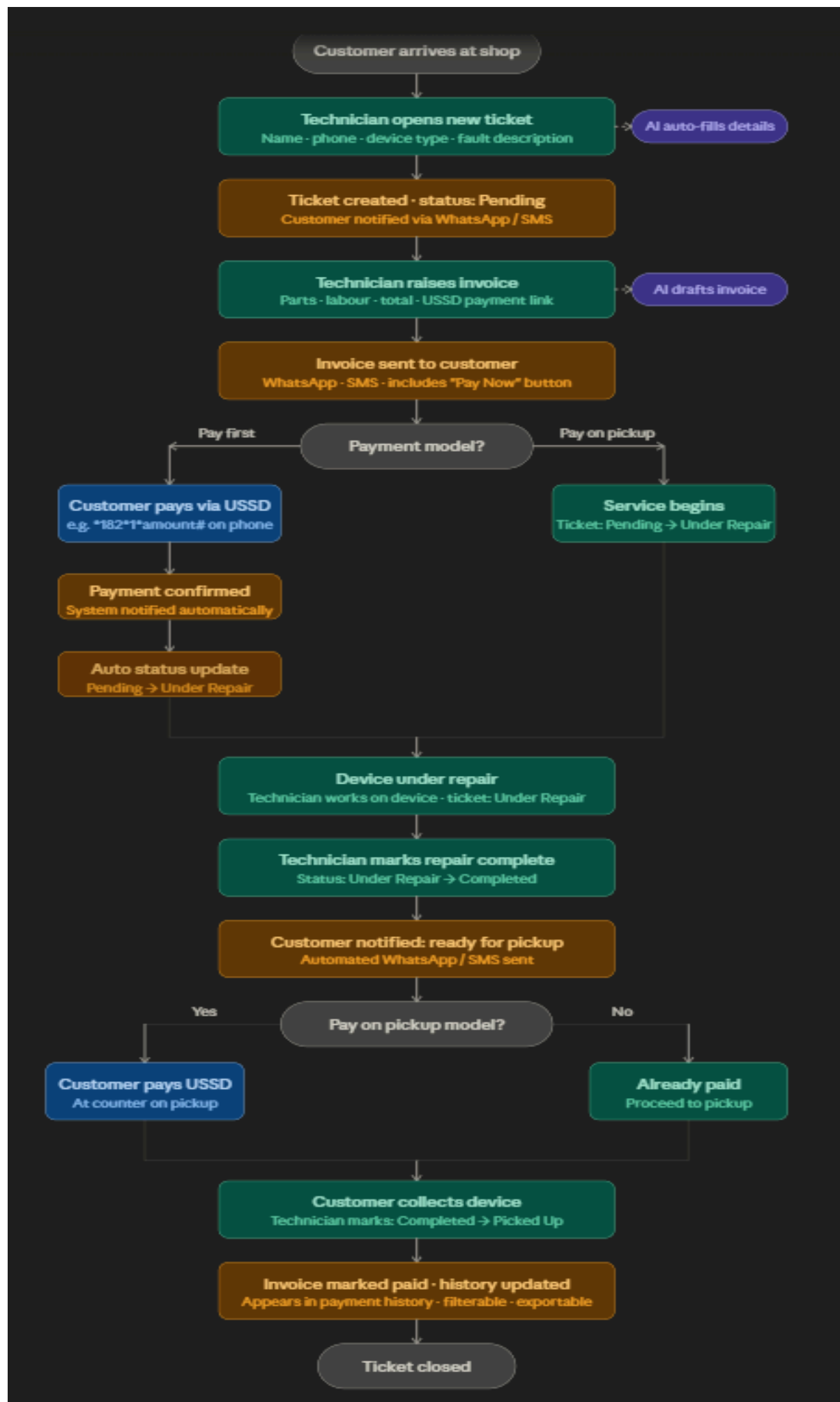
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Once payment is confirmed and the ticket is marked Picked Up, the system finalises the transaction.

- Invoice status updated to: Paid
- Transaction appears in payment history
- Admin can filter, view, and export records



Ticket closed — repair cycle complete



TICKET STATUSES

Status Lifecycle

Every repair ticket moves through the following statuses. Status changes trigger automated customer notifications where indicated.

Status	Meaning	Triggered by
Pending	Ticket created, awaiting payment or start of service	System (auto on ticket creation)
Under Repair	Payment confirmed or technician has begun work	Auto (after USSD payment) or Manual (technician)
Completed	Device has been fully repaired	Technician (manual update)
Picked Up	Customer has collected their device	Technician (manual update)

ADMIN CAPABILITIES

What the Admin can do at any time

In addition to the technician flow above, the admin has access to the following capabilities from their dashboard at any point.

Capability	Description
Manage team	Add or remove technician accounts and manage user permissions
Invoice management	View full invoice list, filter by date range, status, or technician
Payment history	Review all payments received over any time period with filter controls
Export records	Download invoice or payment history as CSV or PDF
Revenue dashboard	Summary of total invoiced, total received, and outstanding amounts
Language toggle	Switch the entire interface between English (EN) and Kinyarwanda (RW)

USSD Payment — How it works

StackFix uses USSD (Unstructured Supplementary Service Data) as its primary payment method. This was chosen specifically for the Kigali, Rwanda market.

Why USSD?

- Works on all phones including basic feature phones
- No internet connection required to pay
- Familiar to Rwandan users (MoMo, Airtel Money)
- No complex card integrations needed at launch
- Instant confirmation sent to both customer and system

How it works

- Customer receives invoice with a USSD code (e.g. *182*1*amount#)
- Customer dials the code on their phone
- They confirm the payment using their mobile money PIN
- System receives automated payment confirmation
- Ticket status updates automatically
- Card payments may be added in a future version

AI FEATURES

AI Assistance in StackFix

StackFix includes an AI assistant designed to reduce manual input and speed up the technician's workflow. The AI does not replace the technician — it assists them.

- Auto-suggest device type and model based on partial input
- Auto-suggest common fault descriptions from a trained library
- Draft invoice line items based on device and fault type (e.g. Samsung A54 + Screen crack → suggests Screen Replacement + Labour)
- Flag incomplete ticket information before submission
- Smart search: find past tickets by customer name, device, or fault type